Q.1 {{\text{In polar coordinate system, x - axis is also called}}}

A. {{{None of these}}}

B. {{{Imaginary axis}}}

C. {{{Polar axis}}}

D. {{{Pole}}}

Q.2 {\text{The}}\,\,{\text{curve}},\,\,r = \theta \,\,(\theta \geqslant 0),\,\,{\text{is}}\,\,{\text{an}}\,\,{\text{equation}}\,\,{\text{of}}\,\,_____.

A. {{cardioids}}

B. {{limacons}}

C. {{roses}}

D. {{spiral}}\,\,{{with}}\,\,a = 1

Q.3 The rose curve has 2n-equally spaced petals of loops if n is _____.

A. even

B. odd

Q.4 {\text{In the integration in polar coordinates, }}dx\,dy{\text{ is replaced by - - - - - - - - - - }}{\text{.}}

A. d

B. r\,dr\,d

C. dr\,d

D. r\,\,dr

Q.5 \begin{gathered} {\text{The }}\,\,{\text{position }}\,\,{\text{of }}\,\,{\text{the}}\,\,{\text{ limacon }}\,\,{\text{relative }}\,\,{\text{to}}\,\,{\text{ the}}\,\,{\text{ polar}}\,\,{\text{ axis}}\,\,{\text{ depends}}\,\,{\text{ on }}\,\,{\text{whether}}\,\,{\text{ _____ }}\,\, \hfill \\ {\text{appears}}\,\,{\text{ in}}\,\,{\text{ the}}\,\,{\text{ equation }}\,\,{\text{and}}\,\,{\text{ whether}}\,\,\, + {\text{ }}\,\,{\text{or}}\,\,{\text{ }} - \,\,{\text{occurs}}{\text{.}} \hfill \\\ \end{gathered}

A. {{(d)}}\,\,\,{{Both}}\,\,{{(a)}}\,\,{{or}}\,\,{{(b)}}{{.}}

B. (c)\,\,\,

C. (a)\,\,\,

D. (b)\,\,\,

Q.6 \int\limits_0^\pi {\int\limits_0^1 {{r^2}\,\,dr\,\,d\theta } } \,\, = \,\,\, - - - - - - - -

A. { }{5}

B. { }{2}

C. { }{4}

D. { }{3}

Q.7 {\text{Area}}\,\,{\text{of}}\,\,{\text{double}}\,\,{\text{integral}}\,\,{\text{can}}\,\,{\text{be}}\,\,{\text{calculated}} = {\text{_____}}{\text{.}}

A. {1}{{{{area}}\,\,{{of}}\,\,R}}_R {dA}

B. _R {f(x,y)\,dx\,dy}

C. _R {f(x,y)\,dy}

D. _R {dA}

**Q.8 \begin{gathered} {\text{The}}\,\,{\text{lemniscates}}\,\,{\text{are}}\,\,{\text{centered}}\,\,{\text{at}}\,\,{\text{the}}\,\,{\text{origin,
\hfill \\ {\text{depends}}\,\,{\text{on}}\,\,{\text{the}}\,\,{\text{sign}}\,\,{\text{preceding}}\,\,{\text{the}}\,\,{a^2}\,\,{\text
\hfill \\\ \end{gathered}**

A. (a)\,\,\, 2

B. (b)\,\,\, 2

C. {{(d)}}\,\,\,{{Both}}\,\,{{(a)}}\,\,{{or}}\,\,{{(b)}}{{.}}

D. (c)\,\,\,

Q.9 {\text{If}} p(r,\theta) {\text{is a point in polar coordinate system, then}} \theta {\text{is called}}

A. {{Reflex angle of }}p

B. {{Acute angle of }}p

C. {{Reflex angle of }}p

D. {{Polar angle of }}p

Q.10 {\text{The graph of the equation }}r\, = \,a(1 - \cos \,\theta){\text{ is symmetric about - - - - - - - - - }}{\text{.}}

A. {{y - axis}}

B. {{pole}}

C. {{None of these}}{{.}}

D. {{initial line}}
